

KAOLACK, Senegal

WMO: 616790

Lat: 14.147N

Lon: 16.051W

Elev: 26

StdP: 14.68

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	62.7	64.7	29.8	23.9	81.1	33.8	28.4	78.2	20.4	76.6	18.2	79.5	7.5	0	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	30.8	107.7	73.6	105.6	73.2	103.3	73.3	84.5	93.0	83.6	91.9	82.7	90.9	7.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.6	169.9	88.8	81.6	164.6	87.9	80.8	160.1	87.0	48.6	93.4	47.6	92.3	46.6	91.2	94.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	15.3	13.8	DB	56.6	112.1	3.7	1.5	54.0	113.2	51.8	114.1	49.7	114.9	47.1	116.0	(4)
(5)			WB	48.0	86.9	2.3	2.3	46.3	88.6	45.0	89.9	43.7	91.2	42.1	92.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	85.0	79.8	83.0	86.4	87.9	88.2	87.4	85.7	84.1	84.2	86.9	85.5	81.1
	DBStd	4.08	3.66	3.87	4.10	3.81	3.25	2.52	2.65	2.60	2.61	2.75	2.55	3.09
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12787	924	923	1130	1139	1185	1121	1106	1058	1027	1144	1065	965
	CDD65	7311	459	503	665	688	720	671	641	593	577	679	615	500
	CDH74	90123	5076	6128	8795	9224	9141	8356	7824	6836	6771	8751	7664	5557
	CDH80	49780	2738	3640	5642	6055	5592	4445	3670	2845	2988	4659	4451	3055
Wind	WSAvg	6.5	7.2	8.1	7.9	8.3	7.8	7.7	6.2	5.1	4.2	4.6	5.0	6.2
Precipitation	PrecAvg	24.1	0.0	0.0	0.0	0.0	0.1	1.5	5.5	8.7	6.5	2.1	0.2	0.0
	PrecMax	39.0	0.7	1.4	0.4	0.0	1.4	6.2	14.5	19.6	14.7	8.4	3.3	0.7
	PrecMin	12.2	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.6	0.4	0.0	0.0	0.0
	PrecStd	6.3	0.1	0.2	0.0	0.0	0.3	1.4	3.3	3.5	3.0	1.9	0.6	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.5	105.1	108.8	110.2	110.4	105.2	98.8	96.6	96.7	101.8	103.2	100.7
		MCWB	72.1	72.0	71.7	72.7	74.0	76.6	80.8	80.7	83.1	76.8	73.9	71.9
	2%	DB	97.8	102.2	106.1	107.8	107.4	101.0	96.3	93.6	94.5	99.1	100.9	98.7
		MCWB	70.8	71.2	72.0	73.2	74.7	77.9	80.6	81.3	82.8	78.7	73.8	71.2
	5%	DB	95.4	99.9	104.1	105.7	104.2	98.4	94.4	92.1	92.9	97.2	99.3	96.5
		MCWB	69.6	70.3	71.6	72.9	75.1	78.0	80.3	81.3	82.2	79.8	73.1	70.2
	10%	DB	93.0	96.9	101.3	102.7	100.7	95.9	92.6	90.4	91.2	95.4	97.3	93.9
		MCWB	68.1	68.9	71.0	72.9	75.4	78.2	80.0	81.0	81.7	80.2	72.8	69.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.8	80.2	79.9	82.0	82.7	84.2	85.2	85.0	85.4	85.6	82.5	79.1
		MCDB	94.7	97.1	101.6	101.1	100.5	95.5	92.9	91.3	93.0	94.2	94.7	94.4
	2%	WB	74.8	75.5	77.1	78.9	80.5	82.1	83.5	83.6	84.2	84.1	80.4	75.7
		MCDB	93.7	96.5	99.8	99.9	96.3	93.6	91.5	90.4	91.5	93.2	91.0	91.4
	5%	WB	72.2	73.3	74.9	76.7	79.0	80.8	82.2	82.6	83.2	83.1	78.9	73.5
		MCDB	91.5	94.7	97.4	97.3	94.4	92.2	90.1	89.4	90.4	91.6	89.0	90.9
	10%	WB	69.8	71.1	72.7	74.8	77.6	79.7	81.2	81.7	82.3	82.1	77.5	71.4
		MCDB	89.7	92.7	94.3	95.5	92.6	90.7	89.0	88.1	89.3	90.2	88.9	88.3
Mean Daily Temperature Range	5% DB	MDBR	26.9	28.2	29.9	30.8	27.1	20.0	15.0	13.2	14.2	18.4	26.0	26.6
		MCDBR	30.0	31.3	32.7	32.9	31.7	23.4	17.3	15.5	16.1	20.9	28.3	30.3
		MCWBR	14.9	13.5	11.6	10.7	9.5	7.1	5.8	5.8	6.6	7.3	10.5	13.2
	5% WB	MCDBR	28.8	29.6	30.7	30.0	26.0	20.0	15.5	14.3	15.4	17.7	24.4	27.6
		MCWBR	16.7	15.1	13.4	12.2	9.3	7.6	7.0	6.7	7.3	7.4	9.9	14.1
Clear-Sky Solar Irradiance	taub	0.519	0.563	0.655	0.676	0.716	0.733	0.669	0.600	0.581	0.604	0.526	0.496	
	taud	1.870	1.769	1.590	1.607	1.537	1.584	1.727	1.890	1.922	1.846	1.944	1.969	
	Ebn at Noon	246	241	223	219	206	200	214	233	238	228	242	248	
	Edh at Noon	62	72	88	86	91	86	75	64	62	66	57	55	
All-Sky Solar Radiation	RadAvg	1682	1881	2129	2245	2200	2047	1845	1749	1745	1821	1693	1603	
	RadStd	82	96	78	73	80	51	66	70	79	73	80	74	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page